Theory in research

A theory can be defined as:

A theory is a set of interrelated constructs (concepts), definitions, and propositions that present a systematic view of phenomena by specifying relations among variables, with the purpose of explaining and predicting the phenomena.

This definition says three things: (1) a theory is a set of propositions consisting of defined and interrelated constructs, (2) a theory sets out the interrelations among a set of variables (constructs), and in so doing, presents a systematic view of the phenomena described by the variables, and (3) a theory explains phenomena; it does so by specifying which variables are related to which variables and how they are related, thus enabling the researcher to predict from certain variables to certain other variables.

Basically, theory helps provide a framework by serving as the point of departure for the pursuit of a research problem. Theory identifies the crucial factors. It provides a guide for organizing and connecting the various facets of the research.

However, besides providing the systematic view of the factors under study, the theory may also very well identify gaps, weak points, and inconsistencies that indicate the need for additional research. Also, the development of the theory may light the way for continued research on the phenomena under study. In educational research, theory serves as synthesizing function, combining ideas and individual bits of empirical information into a set of constructs that provides for deeper understanding, broader meaning and wider application.

The theory aids in defining the research problem; that is, it helps to identify the proper questions to be asked in the context of the specific project.

A theory also provides one or more generalizations that can be tested and then used in practical applications and further research. This development of generalizations is based on the assumption that generalizations do exist in education (or in any area under study) and that individual observations are special cases of such generalizations.

Purpose of a theory

First, theory summarizes and puts in order the existing knowledge in a particular area. It permits deeper understanding of data and translates empirical findings into a more easily retainable and adaptable form. The theory of oxidation for instance, places into focus many of the chemical reactions common to everyday life.

Secondly, theory provides a provisional explanation for observed events and relationships. It identifies the variables that are related and the nature of their relationships. A theory of learning, for example, could explain the relationship between the speed and efficiency of learning and such other variables as motivation, reward and practice.

Lastly, theory permits the prediction of the occurrence of phenomena and enables the investigator to postulate and, eventually, to discover hitherto unknown phenomena. At the time when the 'Periodic Table' was being completed, for instance, certain gaps were noted in the sequence of the elements. Since theory provides that, there should have been no gaps, scientists were spurred on to look for the other missing elements. In time, these were found, anticipated by theory. Theory, therefore, stimulates the development of new knowledge by providing the lead for further inquiry.

Characteristics of a Good Theory

Conditions under which research is conducted and data are obtained within and across studies tend to be more valuable when incorporated into a meaningful unit: standing alone, they may not mean much. As the facts of the research study, the data derive significance from the theory or theories which they fit. Conversely, the theories become acceptable to the extent that they enhance the meaning of the data. Through this process, more adequate theories and unobstructed facts are secured; theory stimulates research and conversely, research stimulates theory development and theory testing.

The criterion by which we judge a theory is not its truth or falsity, but rather its usefulness. Theories sometimes decrease in usefulness in the light of new knowledge, and they are combined, replaced and refined as more knowledge is made available.

A good theory is developed in such a way that the generalisations can be tested. The theory must be compatible with the observations made relative to it and with already existing knowledge. It must adequately explain the events or phenomena under study. The greater the generalisability of the theory the more useful it will be because of its wider applicability.

Another characteristic of a good theory is reflected in the law of parsimony, which holds that a theory should be stated in the simplest form that adequately explains the phenomena. This does not mean that all theories should be simple statements; rather, they should be stated succinctly and precisely, avoiding ambiguities and unnecessary complexity. Important factors must not be overlooked, and the comprehensives of the theory must be adequate for its purpose.

THEORETICAL FRAMEWORK

A theoretical framework comprises the theories expressed by experts in the field into which you plan to research, which you draw upon to provide a theoretical coat hanger for your data analysis and interpretation of results. Put differently, the theoretical framework is a structure that

summarizes concepts and theories, which you develop from previously tested and published knowledge which you synthesize to help you have a theoretical background, or basis for your data analysis and interpretation of the meaning contained in your research data.

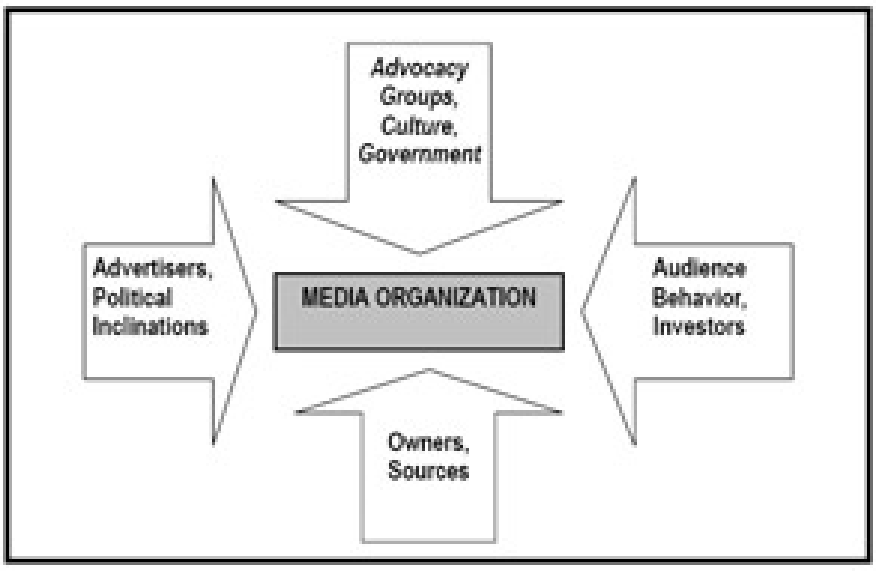
The theoretical framework for your research proposal or project is not a summary of your own thoughts about your research. Rather, it is a synthesis of the thoughts of giants in your field of research, as they relate to your proposed research or thesis, as you understand those theories, and how you will use those theories to understand your data. In essence, the theoretical framework comprises what leaders in your field of research say about your research question, about the problem you plan to investigate, and might even include suggestions of how to solve that problem, including how to interpret the findings in your data. What those leaders say, helps you to develop an informed, and specialized lens, through which you examine your data, conduct the data analysis, interpret the findings, discuss them, and even make recommendations, and conclusions.

The theoretical framework of the study is a structure that can hold or support a theory of a research work. It presents the theory which explains why the problem under study exists. Thus, the theoretical framework is but a theory that serves as a basis for conducting research.

For example:

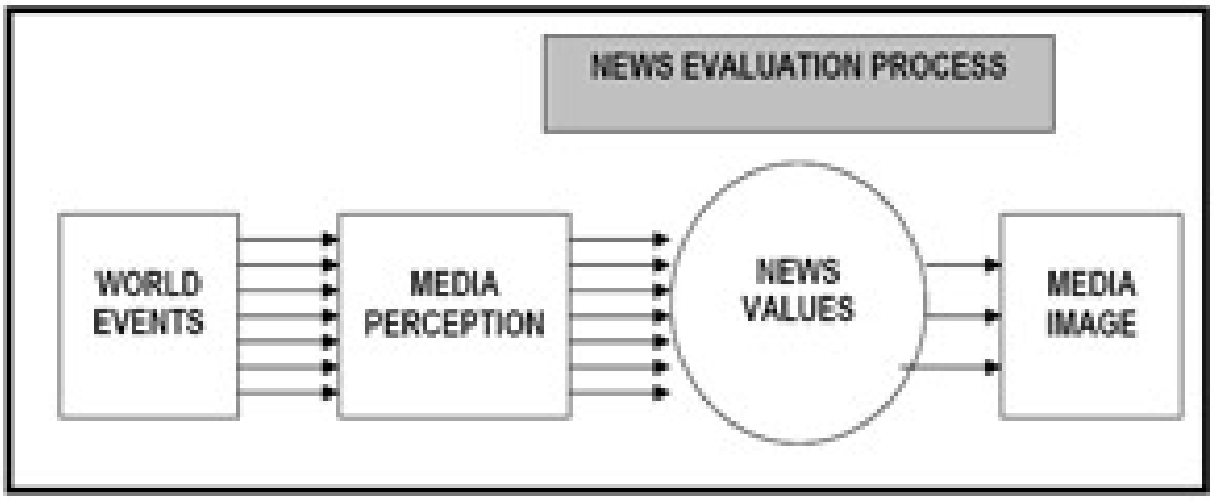
Thesis examines the gatekeeping, news selection and reportage of world news in leading Philippine dailies and primetime newscasts.

Theory 1



Media Organization, Selection, and Production**. McQuail's theory on media as an institution takes into account all forces that affect media's performance**

Theory 2



Selective Gatekeeping. Galtung and Ruge selective gatekeeping theory suggests that news from around the world are evaluated using news values to determine their newsworthiness. (After Mc Quail and Windahl 1993, p. 166)

Formulation of a theoretical framework

1. specifies the theory used as basis for the study
2. mentions the proponents of the theory
3. cites the main points emphasized in the theory
4. Supports his exposition of the theory by ideas from other experts;
5. illustrates his theoretical framework by means of a diagram; and,
6. Reiterates his theoretical proposition in the study. (cralifies)

Asking yourself the following questions, should help you to develop an effective theoretical framework, tailored to your own data analysis needs:

- Is there a theory that can help me to make sense of the meaning of the data that I will gather to answer the research question/s?
- Are there theories that have been developed in the field of my research topic, or in similar topics that might inform an understanding of my research question, my research problem and data analysis?

- What do experts in the field of my proposed research say about the problem I want to investigate?
- What do they say about the research questions I want to investigate, from theoretical perspectives?
- What do they say are the key variables to be investigated?
- Which assumptions, definitions, and propositions are in these leading scholars' theories, and how can I make them explicitly relevant to my research question, research problem, and data analysis?
- How can I modify what they have said, so as to come up with my own theorization about the meanings contained in my research data?
- Have I defined the key concepts in my theoretical framework?
- Does the theoretical framework I am developing address the research questions and hypotheses of my research? (i.e. If you have developed hypotheses).
- Does the theoretical framework I am developing address my research problem?
- Have I included in my theoretical framework all the key theories and analytic models in literature that relate to my study?
- Will the use of this theoretical framework enable me to analyze and discuss the findings in my data?
- Does the theoretical framework have a logical structure and make academic sense?
- Is my theoretical framework easily applicable to my data analysis?
- Does my theoretical framework encompass all the key variables or factors in my data?
- Does my theoretical framework enable me to explain the meaning contained in my data?
- Have I used the latest data available on the theoretical aspects of my planned research?
- Given my theoretical framework, which data shall I include in the analysis, and which data shall I exclude, and why?

Some research utilizes models to provide an overall framework for how we look at reality, approaching in different ways its ontology (nature of being) and epistemology (investigation of what distinguish justified belief from opinion). Examples of models in social research are:

Functionalism: looks at the functions of social institutions;

• Behaviorism: defines all behavior in terms of "stimulus" and "response";

• Interactionism: studies how we attach symbolic meanings to interpersonal relations;

• Ethnomethodology: encourages the researcher in looking at people's everyday ways of producing orderly social interaction.

CONCEPTUAL FRAMEWORK

Concepts

A **concept** is an image or symbolic representation of an abstract idea. Chinn and Kramer (1999) define a concept as a "complex mental formulation of experience".

A conceptual framework is the total, logical orientation and associations of anything and everything that forms the underlying thinking, structures, plans and practices and implementation of your entire research project. So, the conceptual framework comprises your thoughts on identification of the research topic, the problem to be investigated, the questions to be asked, the literature to be reviewed, the theories to be applied, the methodology you will use, the methods, procedures and instruments, the data analysis and interpretation of findings, recommendations and conclusions you will make Thus, the conceptual framework is the logical conceptualization of your entire research project. Saying that it is a logical conceptualization means that a conceptual framework is a metacognitive, reflective and operational element of the entire research process. This in turn means that the conceptual framework involves high order consideration of the following questions and/or issues about your research:

1. What do you want to do in your research? For example, as articulated in your research topic.

2. Why do you want to do it? For example, why is it important to conduct that research? Why it is significant? What aims will it seek to achieve? Which specific objectives will it pursue? How much scope will it cover?

3. How do you plan to do it? For example, which methodology will you apply? Which methods will you use? Who will be your participants? How will you gather data? How will you analyse the data?

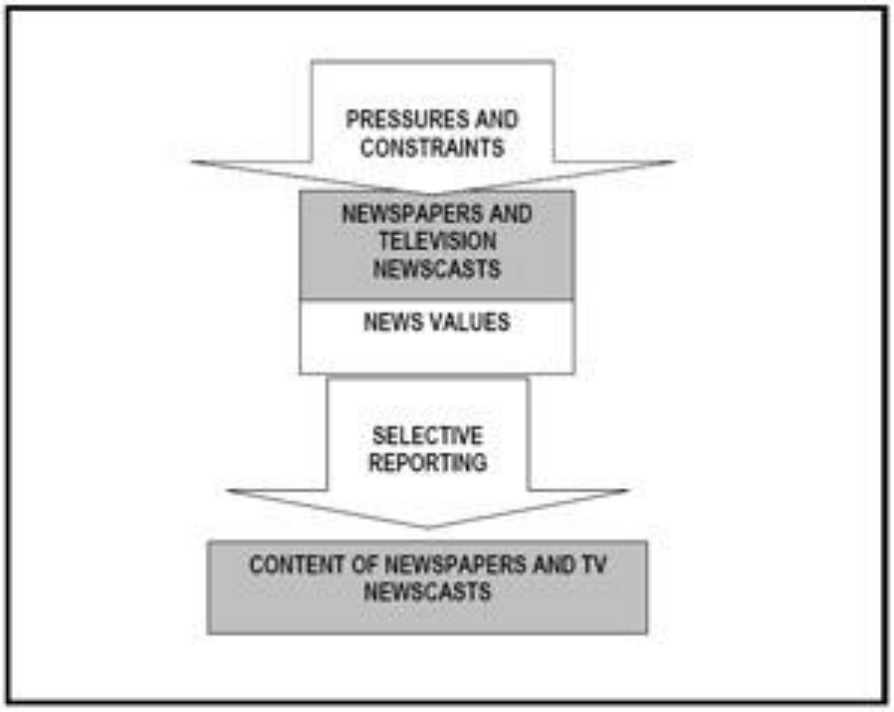
4. How will you make meaning of the data? For example, which theoretical framework will you use to analyse the data? Which software will you use? Which skills will you need?

5. Which worldview will you locate your research in? For example, will it be in the positivist or interpretivist paradigm; critical or pragmatic paradigm?

6. How will you report your findings? For example, in a research paper, or a seminar paper, or a conference paper, a book chapter, or a book or a thesis?

In other words, the conceptual framework is the researcher's understanding of how the particular variables in his study connect with each other. Thus, it identifies the variables

required in the research investigation. It is the researcher's "map" in pursuing the investigation.

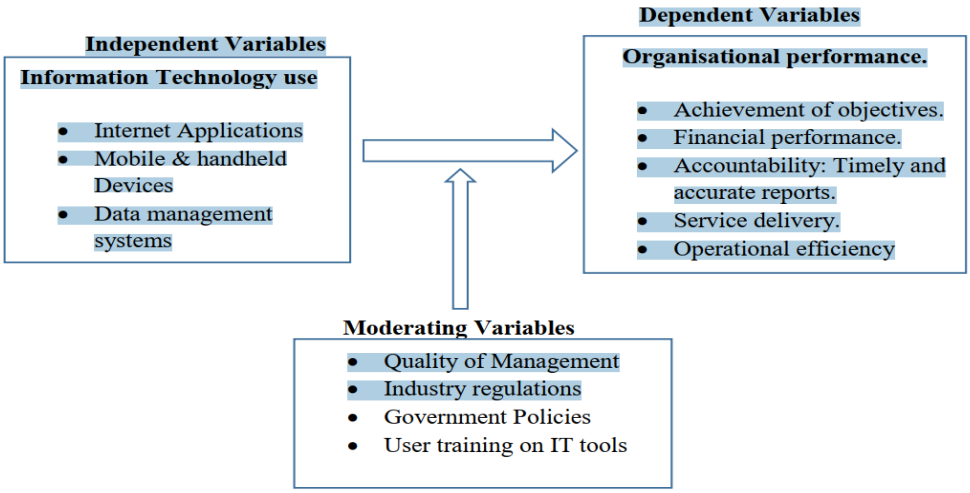


Conceptual Framework. News values, and the pressures and constraints shape the newspapers and television newscasts. The result of this relationship is selective reporting of events.

Formulating a conceptual framework

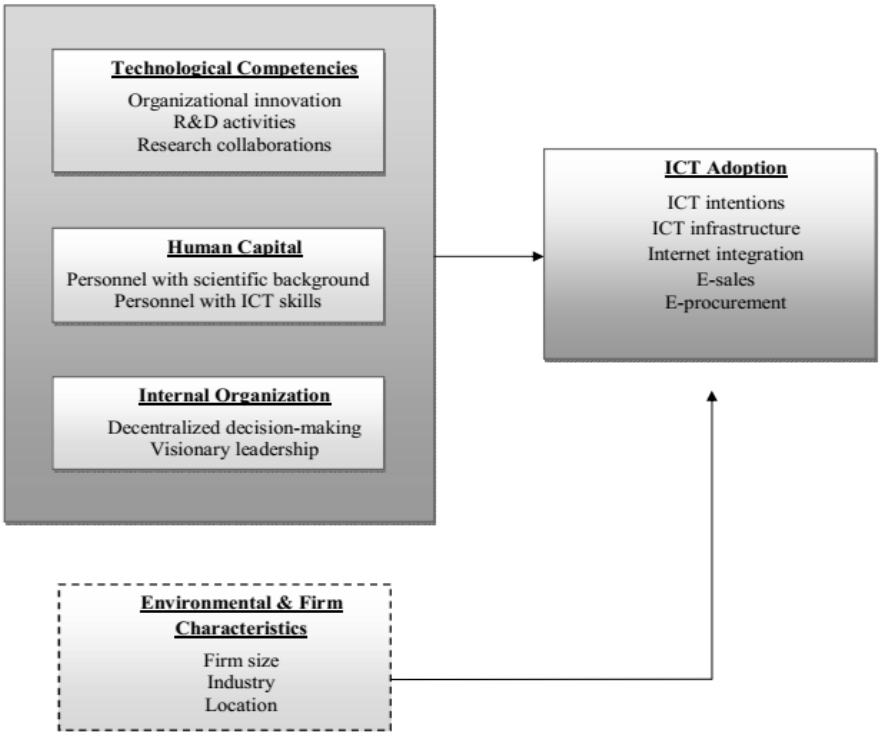
1. cite your conceptual framework or paradigm;
2. Identify your variables;
3. Point out the dependent and intervening variables;
4. Show the direction of the study.

It can be presented in the form of; flow diagrams, tree diagrams, mind maps or even shape based diagrams. Below are some examples for better understanding:

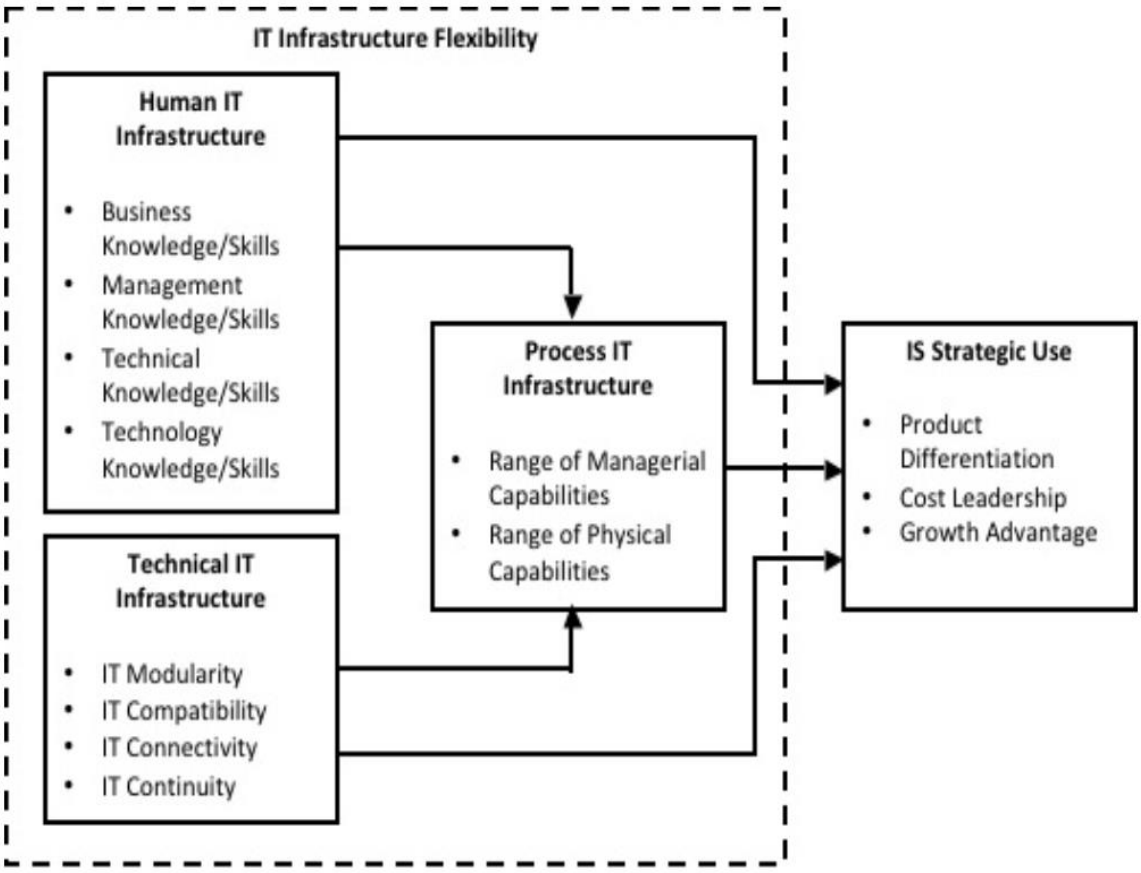


A **moderator variable**, commonly denoted as just M, is a third variable that affects the strength of the relationship between a dependent and independent variable In correlation, a moderator is a third variable that affects the correlation of two variables. In a causal relationship, if x is the predictor variable and y is an outcome variable, then z is the moderator variable that affects the casual relationship of x and y. Most of the moderator variables measure causal relationship using regression coefficient. The moderator variable, if found to be significant, can cause an amplifying or weakening effect between x and y.

What drives ICT adoption by SMEs? Evidence from a large-scale survey in Greece



A Conceptual Framework on IT Infrastructure Flexibility as Determinant of Strategic Utilization of Information Systems


IT Infrastructure Flexibility
Human IT Infrastructure
Business Knowledge/Skills
Management Knowledge/Skills
Technical Knowledge/Skills
Technology Knowledge/Skills
Technical IT Infrastructure
IT Modularity
IT Compatibility
IT Connectivity
IT Continuity
Process IT Infrastructure
Range of Managerial Capabilities
Range of Physical Capabilities
IS Strategic Use
Product Differentiation
Cost Leadership
Growth Advantage